

AVERAGE

$$\text{Average} = \frac{\text{Sum of Observations}}{\text{No. of Observations}}$$

Average of an Arithmetic Progression

- i) Middle term
- ii) (First term + Last term)/2

Arithmetic Mean of 'a' and 'b' = $\frac{a+b}{2}$

Geometric Mean of 'a' and 'b' = $\sqrt{a * b}$

Harmonic Mean of 'a' and 'b' = $\frac{2ab}{(a+b)}$

Sum of Series:

$$1+2+3+4+\dots+n = \frac{n(n+1)}{2}$$

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$$

EXERCISE

1. Find the average of: 8, 12, 16, 20, and 24.

2. Find the average of: 11, 13, 15, 17, 19, and 21.

3. Find the average of: 32, 37, 39, 43, 47 and 54.

4. Find the average of: 56, 61, 63, 69, 78, 81 and 82.

5. Find the average of first 30 natural numbers.

- (a) 30 (b) 31
- (c) 15 (d) 15.5

6. Find the average of first 30 even numbers.

- (a) 30 (b) 31
- (c) 15 (d) 15.5

7. Find the average of first 30 odd numbers.

- (a) 30 (b) 31
- (c) 15 (d) 15.5

8. The average of 9 consecutive numbers is 40. The largest of these numbers is?

- (a) 43 (b) 42
- (c) 44 (d) 41

9. 14 consecutive numbers are given. If the average of the two numbers that appear in the middle is 6.5 then, the sum of these given numbers is?

- (a) 91 (b) 84
- (c) 78 (d) 104

10. The average of 25 consecutive odd integers is 55. The highest of these integers is?

- (a) 79 (b) 105
- (c) 109 (d) 155

11. Average weight of 17 students is 16 kg. A new student joins and the average becomes 21 kg. Find the weight of the new player.

- (a) 100kg (b) 102kg
- (c) 104kg (d) 106kg

12. Average weight of 25 students of a class is 50 kg. If the weight of the class teacher is included, the average is increased by 1 kg. The weight of the teacher is

- (a) 76 kg (b) 75 kg
- (c) 77 kg (d) 74 kg

13. Average age of 17 students of a class is 23 years. One student left the class, now the average becomes 24 years. Find the age of that student.

- (a) 41 years (b) 17 years
- (c) 7 years (d) 8 years

14. A player scores 92 runs in his 21st innings after that his average increases by 2 runs. Find the average of the player after 21st innings.

- (a) 52 (b) 50
- (c) 48 (d) 46

15. The average of nine numbers is 20. One of the numbers is 21 and if it is replaced by 30 then, the average becomes?

- (a) 21.5 (b) 19.5
- (c) 21 (d) 20

16. The average weight of the 10 people in a boat is increased by 1.2 kg when one of the crew members who weigh 65kg is replaced by a new man. The weight of the new man (in kg) is?

- (a) 70 kg (b) 68 kg
- (c) 71 kg (d) 77 kg

17. The average of ten numbers is calculated as 17. It was discovered later on that while calculating the average one number namely 49 was wrongly read as 39. Find the correct average.

- (a) 16 (b) 18
(c) 26 (d) 20

18. The average marks of 45 students was found out to be 72. If the marks of two students were incorrectly entered as 58 and 62 instead of 85 and 26 respectively then what is the correct average?

- (a) 72.2 (b) 67.9
(c) 68.4 (d) 71.8

19. Average of the three numbers is 77. The 1st no. is twice the 2nd and the 2nd is twice the third. The first no. is?

- (a) 33 (b) 66
(c) 126 (d) 132

20. There are 5 numbers. The average of first four is 26 and the average of last four is 25. What will be the difference between first and the last number?

- (a) 1 (b) 2
(c) 3 (d) 4

21. The average temperature for Monday, Tuesday, Wednesday and Thursday was 48° . The average temperature for Tuesday, Wednesday, Thursday and Friday was 52° . If the temperature on Monday was 42° then, the temperature on Friday was (in degrees)

- (a) 58 (b) 56
(c) 52 (d) 50

22. 9 people go to a restaurant, 8 out of them spent Rs.25 each. 9th person spends Rs.40 more than the average amount of all the 9 persons. Find the total expense of the dinner.

- (a) 300 (b) 245
(c) 270 (d) 450

23. There are 35 students in a hostel. 7 new students join the hostel. Now, per day expenditure of mess was increased by Rs.42 but per head expenditure of mess decreases by Rs.1 find the original expenditure of mess.

- (a) 420 (b) 450
(c) 360 (d) 540

24. Average weight of 57 boys is 90kg. Average weight of 76 girls is 97kg. Find the average weight of the class.

- (a) 92 (b) 93
(c) 94 (d) 95

25. The average weight of a class of 40 students is 30kg and the average weight of another class with 20 students is 15kg. Find the average weight of both the classes combined.

- (a) 35Kg (b) 20kg
(c) 30kg (d) 25kg

****.** Average of 18 numbers is 37.5. If 6 numbers of average 'x' are added to them then, the average of all the numbers is increased by one. The value of 'x' is?

- (a) 38.5 (b) 39.5
(c) 40.5 (d) 41.5
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